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Homogeneous, Second order ODEs: Repeated Roots

$$y(t) = \underbrace{v(t)}_{\text{unknown}} \underbrace{y_1(t)}_{\text{known}}$$

$$y' = v y_1' + y_1 v'$$

Example 3.17 $y'' - y' + \frac{y}{4} = 0 \qquad y(0) = 2 \quad y'(0) = 1/3$

1/ $y = e^{mt} \qquad m^2 - m + \frac{1}{4} = 0 \Rightarrow (m - 1/2)^2 = 0 \Rightarrow m_1 = m_2 = 1/2$

The general solution is $y(t) = c_1 t e^{\frac{1}{2}t} + c_2 e^{\frac{1}{2}t} \quad \forall t$

2/ IVP: $y(0) = 2 \Rightarrow c_2 e^0 = 2 \Rightarrow c_2 = 2$

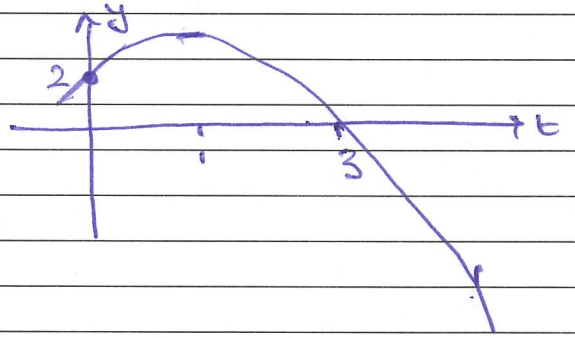
$$y'(t) = c_1 \left\{ t e^{\frac{1}{2}t} \frac{1}{2} + e^{\frac{1}{2}t} \cdot 1 \right\} + c_2 e^{\frac{1}{2}t} \frac{1}{2}$$

$$\Rightarrow y'(0) = c_1 + c_2 \frac{1}{2} = \frac{1}{3} \Rightarrow c_1 = \frac{1}{3} - \frac{1}{2} \cdot 2 = -\frac{2}{3}$$

The solution to the IVP

$$y(t) = -\frac{2}{3} t e^{\frac{1}{2}t} + 2 e^{\frac{1}{2}t} \quad \forall t \;//$$

- $y(0) = 2$
- $y'(0) > 0$
- As $t \to \infty$, $y \to -\infty$
- $\frac{dy}{dt} = \ldots = 0$ @ $t = 1$
- $y = 0 \ldots \Rightarrow t = 3$



In general

- positive exponent; growth in the amplitude (magnitude)
- If negative exponent; decay to zero
- $m_1 = m_2 = 0 \Rightarrow y'' = 0 \Rightarrow$ linear in t.

Example 3.19

$$y'' - 3y' - 4y = 3e^{2t}$$

1/ Homogeneous problem: $y'' - 3y' - 4y = 0$

$y = e^{mt} \qquad m^2 - 3m - 4 = 0 \Rightarrow (m-4)(m+1) = 0$

$\therefore m_1 = -1$ and $m_2 = 4$

$$y_c = c_1 e^{-t} + c_2 e^{4t}$$

2/ $g(t) = 3e^{2t}$: row 2 of table $\alpha = 2 \quad k = 0$

2 is not a root of the characteristic eq^n $\therefore s = 0$

Try $y_1 = B_0 e^{2t} t^0 = B e^{2t}$ and find B.

$y_p' = 2Be^{2t}$ $\quad y_p'' = 4Be^{2t}$

and substitute into the nonhomogeneous problem:

$$4Be^{2t} - 3\cdot 2Be^{2t} - 4Be^{2t} = 3e^{2t}$$

$$\Rightarrow B = -1/2$$

$$y_p = -\frac{1}{2}e^{2t}$$

The general solution is

$$y = y_c + y_p = c_1e^{-t} + c_2e^{4t} - \frac{1}{2}e^{2t} \quad //$$

Example 3.23 $\quad y'' + 4y' + 4y = t^{-2}e^{-2t}$.

1/ nonhomogeneous problem: characteristic eq: $m^2 + 4m + 4 = 0$

$\Rightarrow$ repeated roots $m_1 = m_2 = -2$

Reduction of order gives the fundamental set of solns:

$$y_1 = te^{-2t} \qquad y_2 = e^{-2t}$$

2/ $g = t^{-2}e^{-2t}$ $\qquad W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} te^{-2t} & e^{-2t} \\ te^{-2t}(-2) + e^{-2t} & -2e^{-2t} \end{vmatrix}$

$$= -e^{-4t} \quad (\neq 0)$$

3/ $u_1 = -\int_{s_0}^{t} \frac{y_2 g}{W}\, ds$

$$= -\int \frac{e^{-2s}\, s^{-2}\, e^{-2s}}{-e^{-4s}}\, ds$$

$$= \int s^{-2}\, ds$$

$$= -s^{-1}$$